

Machine Learning (911.236), Statistical Learning Theory (536.505), Advanced Machine Learning (911.936)

Exercise sheet C

Exercise 1.

5 P.

Let $\mathcal{H}_1$ be a hypothesis class of functions $h_{a,b} : \mathbb{R}^2 \rightarrow \{0, 1\}$, defined by

$$\mathcal{H}_1 = \{h_{a,b} : \mathbb{R}^2 \rightarrow \{0, 1\} \mid 0 \leq a \leq b\},$$

where

$$h_{a,b}(x) = 1 \iff a \leq \|x\|_2^2 \leq b.$$

In other words, $\mathcal{H}_1$ consists of all classifiers induced by centered annuli, where the case $a = 0$ is interpreted as a centered disk.

Show that

$$\text{VCdim}(\mathcal{H}_1) = 2,$$

by showing that a set of cardinality 2 can be shattered, while no set of cardinality 3 can be shattered.

Exercise 2.

5 P.

Let $\mathcal{H}_2$ be a hypothesis class of functions $h_{a_1,b_1,a_2,b_2} : \mathbb{R}^2 \rightarrow \{0, 1\}$, defined by

$$\mathcal{H}_2 = \{h_{a_1,b_1,a_2,b_2} : \mathbb{R}^2 \rightarrow \{0, 1\} \mid 0 \leq a_1 \leq b_1 \leq a_2 \leq b_2\},$$

where

$$h_{a_1,b_1,a_2,b_2}(x) = 1 \iff \|x\|_2^2 \in [a_1, b_1] \cup [a_2, b_2].$$

Thus $\mathcal{H}_2$ consists of all classifiers induced by unions of two centered annuli. Degenerate annuli are allowed, and $a_1 = 0$ allows the first annulus to be a centered disk.

Show that

$$\text{VCdim}(\mathcal{H}_2) = 4.$$